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Measuring Head Acceleration Like a CHAMP

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There is a keen interest in developing data-driven strategies to mitigate head injury and reduce repetitive head impacts in sports owing to concerns about potential short- and long-term neurological consequences of these loads to brain. Interventions can include advancements in equipment design, modification to rules of the game and training regimens, and enhancements to clinical assessment after an athlete sustains a concussive impact. One key to driving these improvements in athlete safety is the quantitative biomechanical measurement of head impact kinematics. Over the past 2 decades, technological advances in head kinematic measurement devices have made implementation more feasible and as a result, these devices are sold commercially and used regularly across a range of scientific disciplines including athletic training, exercise/sport science, physical therapy, neuroscience, and biomechanics. In addition, the price point of such devices has been reduced such that they are marketed to and used by parents and teams under the auspices of monitoring and protecting their athletes. A recent systematic review highlighted the exponential increase in peer-reviewed publications using such technology to quantify head loading in athletes and correlate kinematic measures to relevant clinical, physiological, and neuropsychological outcomes.[1]

As use of these devices proliferates, so have concerns about their accuracy and specific methodological challenges that can limit the rigor of data collected.[2–5] Inaccurate or insufficiently/poorly processed data can slow deployment of impactful interventions, limiting research and clinical advances or in the worst-case scenario, increasing neurological risk. As a result, there is a need for methodological guidelines to improve the rigor and consistency of research on head acceleration measurement and reduce the risk of scientific bias. Further, improved research rigor in this area provides a strong evidence base for practitioners as they develop protocols, recommendations and interface with patients, athletes, and families around

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head injury mitigation.

To this end, the Consensus Head Acceleration Measurement Practices (CHAMP) group was founded to develop and recommend best practices for the collection, analysis, and reporting of head acceleration measurement data in sport. Comprised of scientists and clinicians from a variety of backgrounds (including industry representation), the group undertook a comprehensive effort to define current best practices in head kinematic measurement, culminating in a series of manuscripts outlining consensus methodologies.[6–10] CHAMP prioritized 6 areas of focus:

- Study design and statistical analysis in studies of head acceleration measurement
- Laboratory validation of wearable head kinematic devices
- On-field validation and use of wearable head kinematic devices
- Video analysis of head acceleration events
- Physical reconstruction of head acceleration events
- Computational modeling of head acceleration events.

Work groups for each focus area, led by experts in the field, drafted consensus statements that outline currently recommended best practices for many aspects of head acceleration measurement. CHAMP workgroups and key stakeholders convened at a consensus conference held in Philadelphia in March 2022. All attendees at the conference participated in an open scientific discussion of the key concepts and then formally voted on each consensus statement. Details of this process and all participants are summarized here.[11]

Products from this conference include reporting checklists that align with technical manuscripts.[11] The checklists can be referred to as “CHAMP 2022 Reporting Guidelines” and

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provide elements necessary for transparent reporting in peer-reviewed literature for studies using these methods. The CHAMP guidelines are not intended as the final word in head impact measurement reporting, but rather an accessible means for scientists who use head kinematics measurement devices in their research to translate the best practices summarized in the technical manuscripts as they prospectively consider design of their studies and report the results of such work in the peer-reviewed literature. As readers and reviewers of that literature, we all should consider these checklists in our evaluation and interpretation of published work in the field. Other fields have developed checklists outlining reporting recommendations (eg, CONSORT [www.consort-statement.org] and EQUATOR [www.equator- network.org]) and their use has increased completeness of reporting and transparency of research.[12,13]

We are at an exciting time where advances in head acceleration measurement technology have opened tremendous opportunity to apply quantitative biomechanical analyses to head injury mitigation strategies for athletes. Research across multiple disciplines that include such devices will allow the evolution of the sport environment to one where the benefits of athletic participation are realized while minimizing the risk of neurological consequences. Enhanced rigor as outlined by the CHAMP best practices manuscripts and reporting guidelines will ensure that the scientific foundation upon which those advancements are based is strong and robust. We strongly encourage their use.

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